SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
ProgramName:B. Tech		**Assignment Type: Lab**	**AcademicYear:**2025-2026
CourseCoordinatorName		Venkataramana Veeramsetty	
Instructor(s)Name		Dr. V. Venkataramana (Co-ordinator) Dr. T. Sampath Kumar Dr. Pramoda Patro Dr. Brij Kishor Tiwari Dr.J.Ravichander Dr. Mohammand Ali Shaik Dr. Anirodh Kumar Mr. S.Naresh Kumar Dr. RAJESH VELPULA Mr. Kundhan Kumar Ms. Ch.Rajitha Mr. M Prakash Mr. B.Raju Intern 1 (Dharma teja) Intern 2 (Sai Prasad) Intern 3 (Sowmya) NS_2 (Mounika)	
CourseCode	24CS002PC215	**CourseTitle**	AI Assisted Coding
Year/Sem	II/I	**Regulation**	R24
Date and Day of Assignment	Week4 - Wednesday	**Time(s)**	
Duration	2 Hours	**Applicableto Batches**	
AssignmentNumber:8.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*ExpectedTime to complete*
1	Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases **Lab Objectives:** • To introduce students to test-driven development (TDD) using AI code generation tools. • To enable the generation of test cases before writing code implementations.	Week4 - Wednesday

- To reinforce the importance of testing, validation, and error handling.
- To encourage writing clean and reliable code based on AI-generated test expectations.

Lab Outcomes (LOs):
After completing this lab, students will be able to:

- Use AI tools to write test cases for Python functions and classes.
- Implement functions based on test cases in a test-first development style.
- Use unittest or pytest to validate code correctness.
- Analyze the completeness and coverage of AI-generated tests.
- Compare AI-generated and manually written test cases for quality and logic

Task Description#1

Use AI to generate test cases for is_valid_email(email) and then implement the validator function.

Requirements:

- Must contain @ and . characters.
- Must not start or end with special characters.
- Should not allow multiple @.

Expected Output#1

- Email validation logic passing all test cases

Task Description#2 (Loops)

- Ask AI to generate test cases for assign_grade(score) function. Handle boundary and invalid inputs.
  Requirements
- AI should generate test cases for assign_grade(score) where: 90-100: A, 80-89: B, 70-79: C, 60-69: D, <60: F
- Include boundary values and invalid inputs (e.g., -5, 105, "eighty").

Expected Output#2

Grade assignment function passing test suite

Task Description#3

- Generate test cases using AI for is_sentence_palindrome(sentence). Ignore case, punctuation, and spaces

  Requirement
- Ask AI to create test cases for is_sentence_palindrome(sentence)
  (ignores case, spaces, and punctuation).

- Example:
  "A man a plan a canal Panama" → True

Expected Output#3

- Function returns True/False for cleaned sentences
- Implement the function to pass AI-generated tests.

Task Description#4

- Let AI fix it Prompt AI to generate test cases for a ShoppingCart class (add_item, remove_item, total_cost).

  Methods:
  Add_item(name,orice)
  Remove_item(name)

Total_cost()

Expected Output#4

- Full class with tested functionalities

Task Description#5

- Use AI to write test cases for convert_date_format(date_str) to switch from "YYYY-MM-DD" to "DD-MM-YYYY".
  Example: "2023-10-15" → "15-10-2023"

Expected Output#5

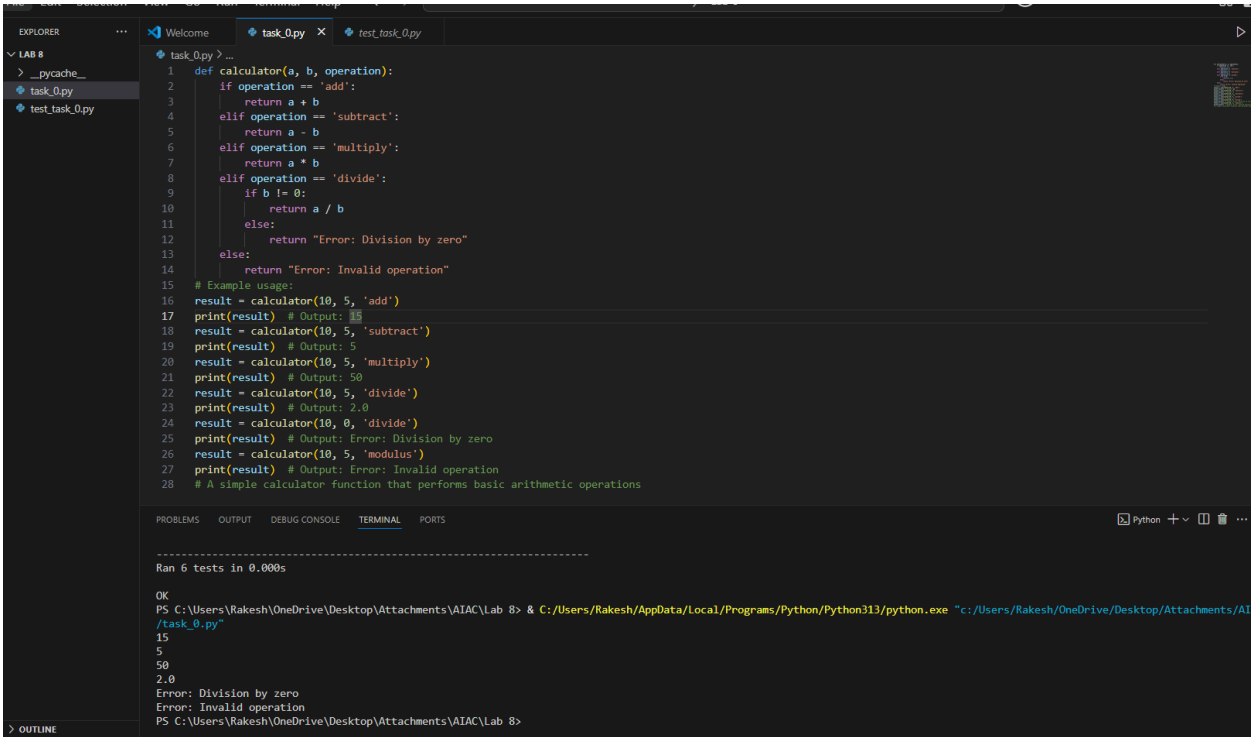
- Function converts input format correctly for all test cases

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

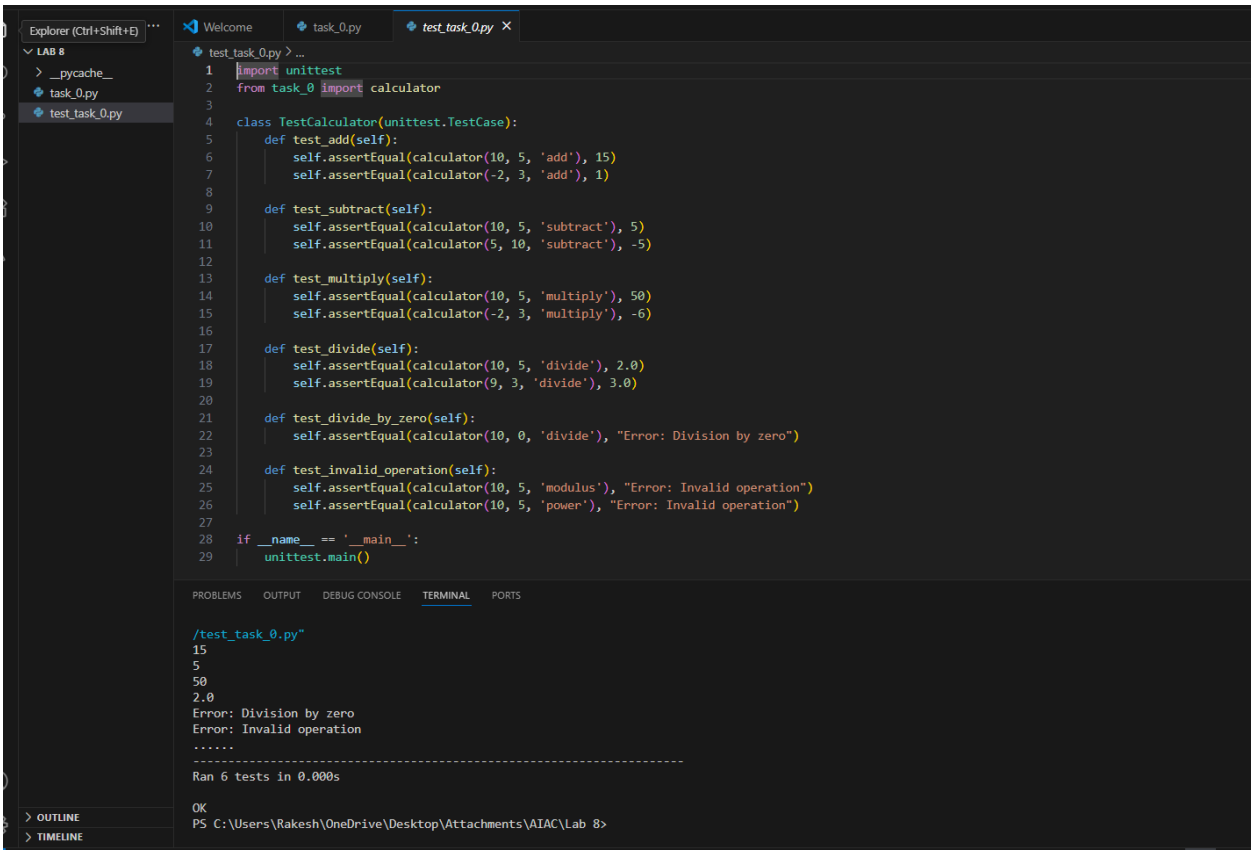
Evaluation Criteria:

Criteria	Max Marks
Task #1	0.5
Task #2	0.5
Task #3	0.5
Task #4	0.5
Task #5	0.5
Total	**2.5 Marks**

TASK 0:



TEST CASE:



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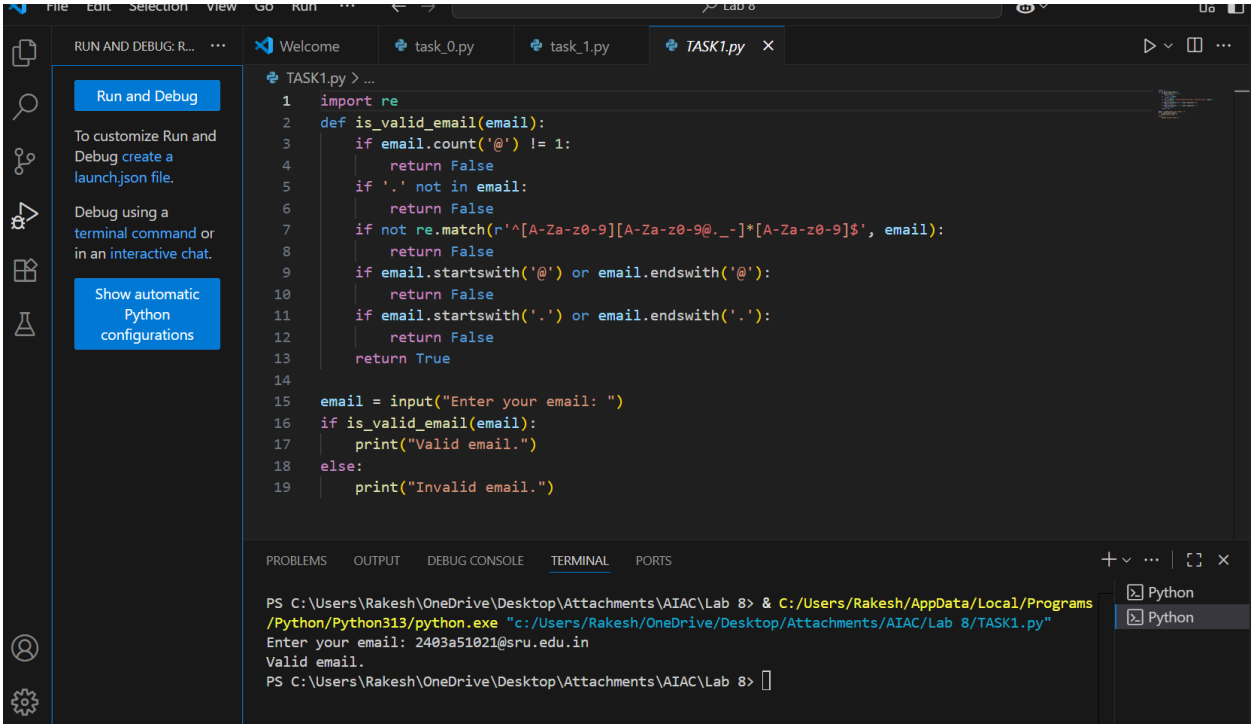
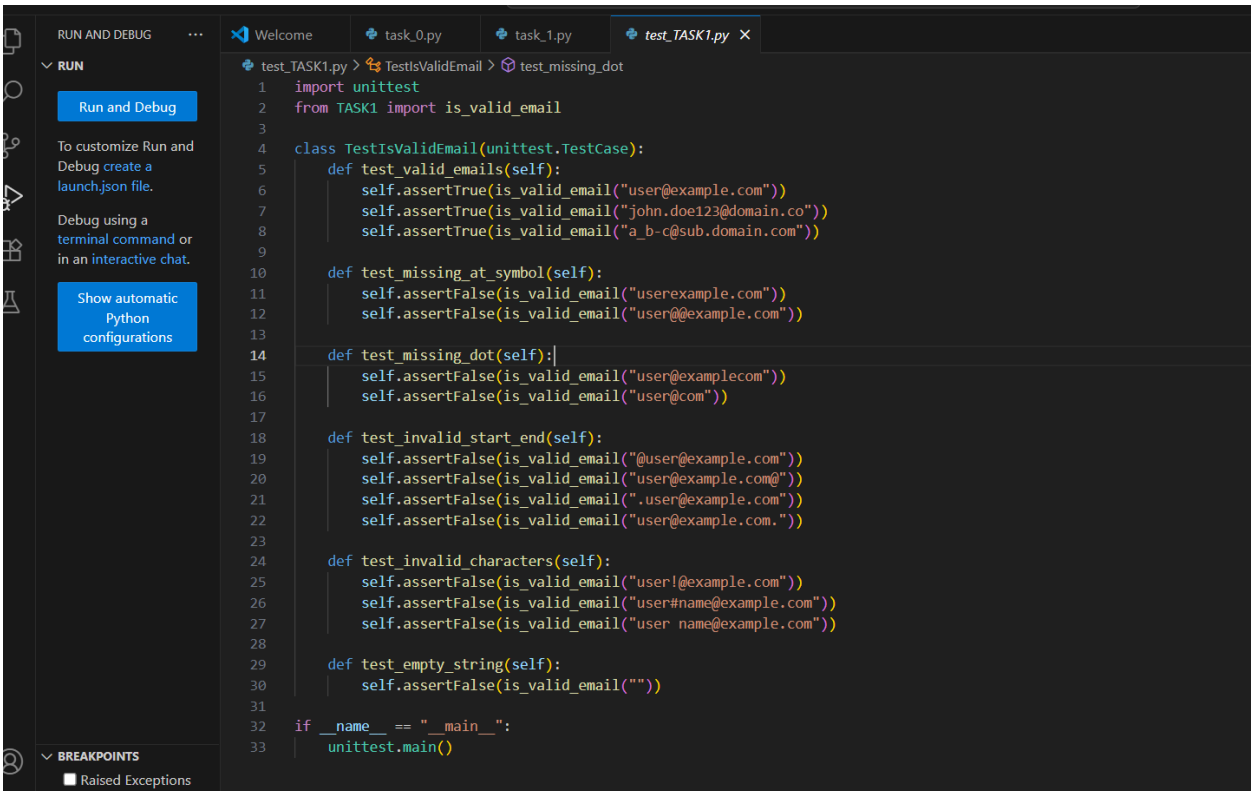
TASK 1:



TEST CASE:



OUTPUT:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Rakesh\OneDrive\Desktop\Attachments\AIAC\Lab 8>  & 'c:\Users\Rakesh\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\Rakesh\.vscode\extensions\ms-python.debugpy-2025.10.0-win32-x64\bundled\libs\debugpy\launcher' '60612' '--' 'C:\Users\Rakesh\OneDrive\Desktop\Attachments\AIAC\Lab 8\test_TASK1.py'
Enter your email: 2403a51021@sru.edu.in
Valid email.
......
----------------------------------------------------------------------
Ran 6 tests in 0.002s

OK
PS C:\Users\Rakesh\OneDrive\Desktop\Attachments\AIAC\Lab 8>
```

Task Description#2 (Loops)

- Ask AI to generate test cases for assign_grade(score) function. Handle boundary and invalid inputs.

Requirements

- AI should generate test cases for assign_grade(score) where: 90-100: A, 80-89: B, 70-79: C, 60-69: D, <60: F
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Expected Output#2

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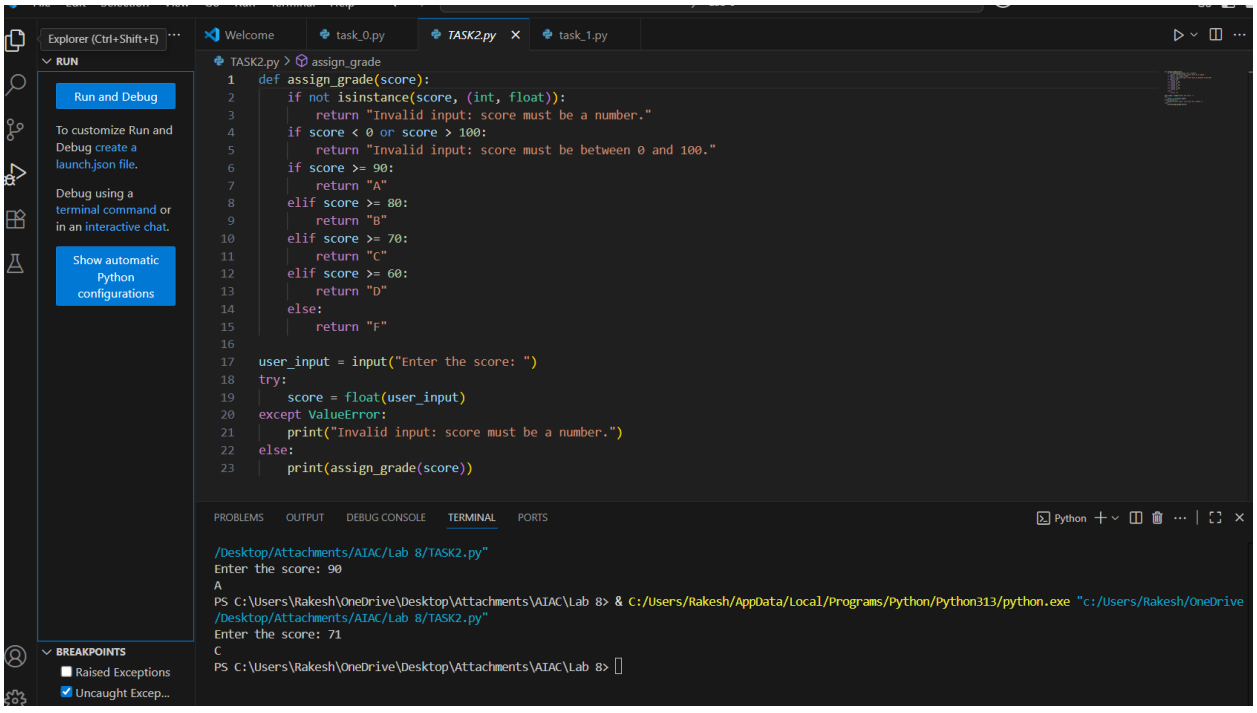
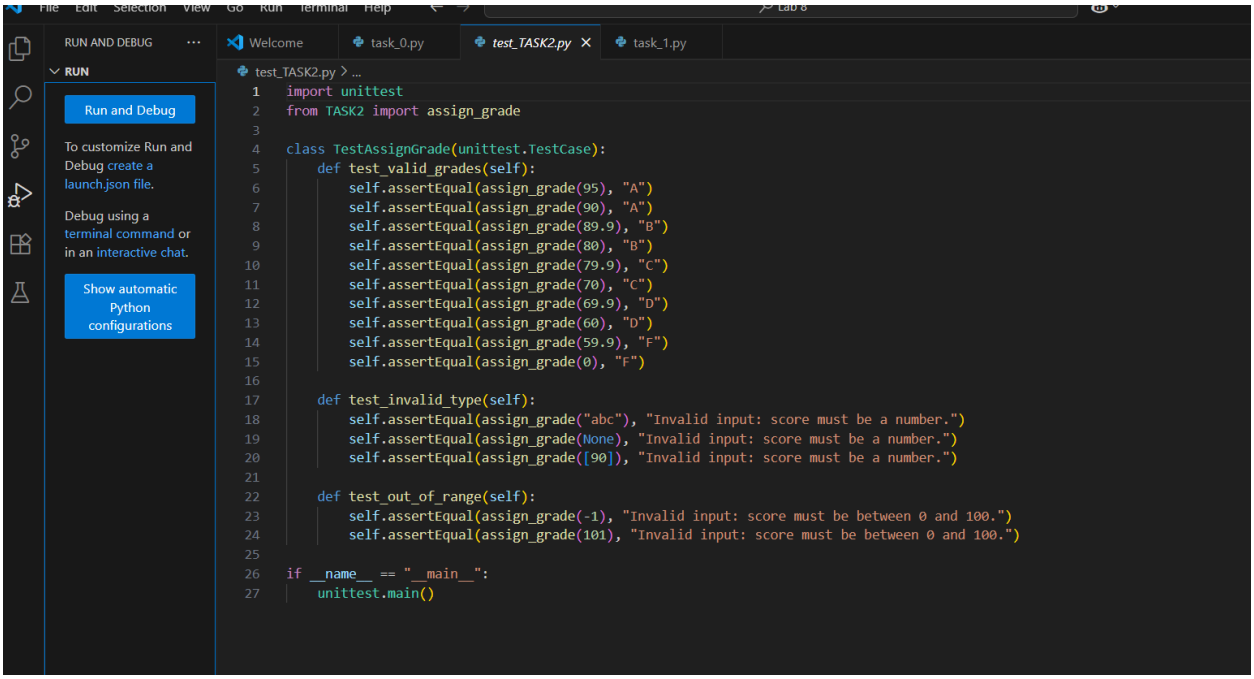
TASK 2:



TEST CASE:



OUTPUT :

Task Description#3

- Generate test cases using AI for is_sentence_palindrome(sentence). Ignore case, punctuation, and spaces

Requirement

- Ask AI to create test cases for is_sentence_palindrome(sentence)
  (ignores case, spaces, and punctuation).

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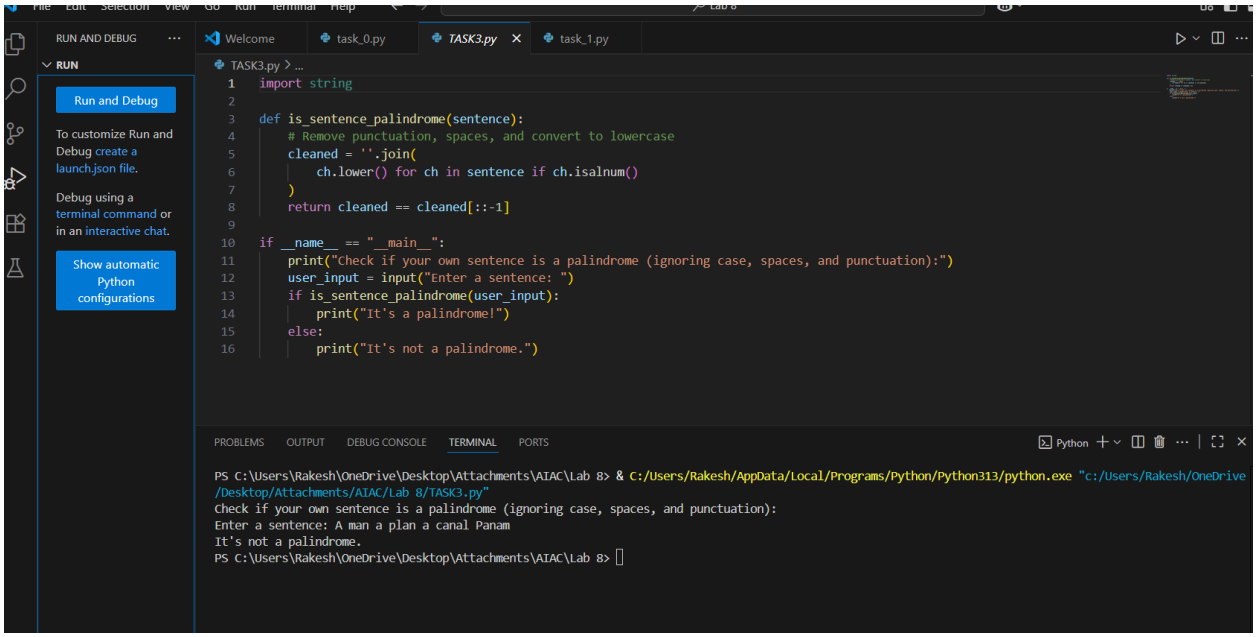
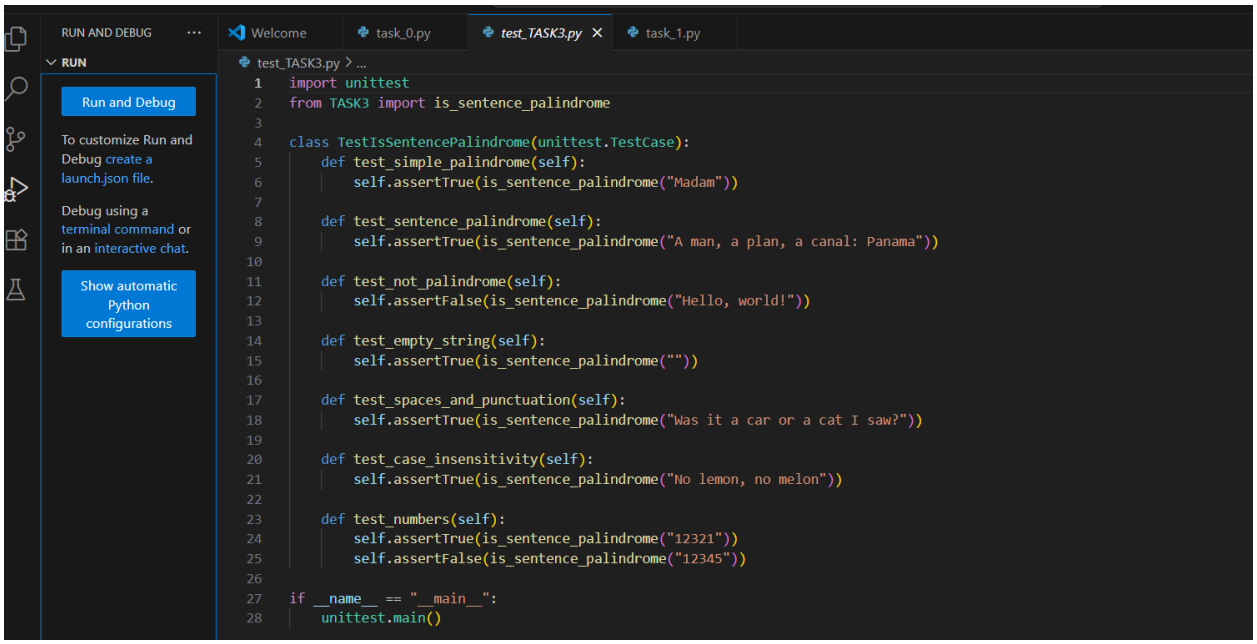
Expected Output#3

- Function returns True/False for cleaned sentences
- Implement the function to pass AI-generated tests.

TASK 3:



TEST CASE:



OUTPUT:

Task Description#4

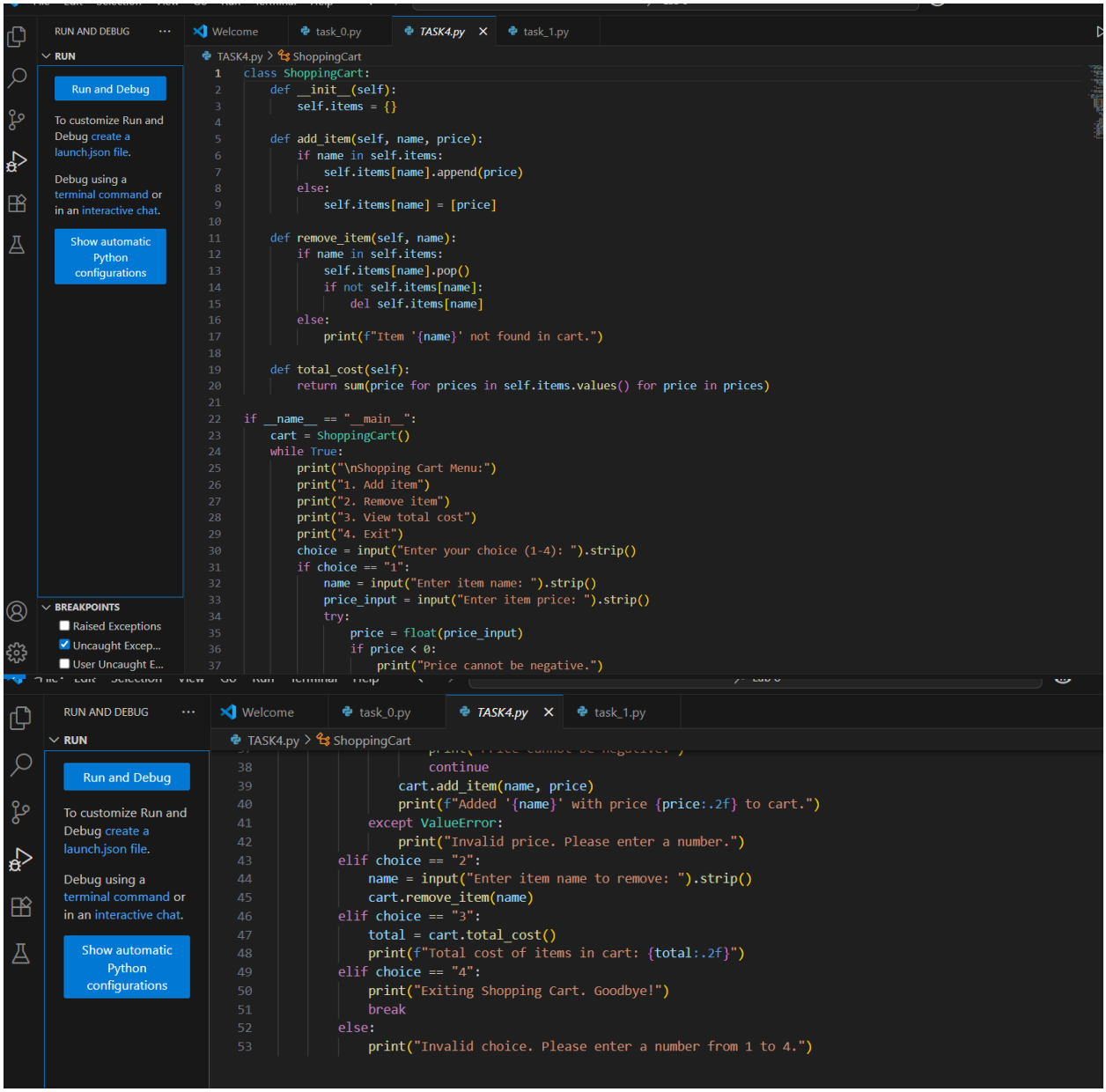
- Let AI fix it Prompt AI to generate test cases for a ShoppingCart class (add_item, remove_item, total_cost).

Methods:
Add_item(name,orice)
Remove_item(name)
Total_cost()

Expected Output#4

- Full class with tested functionalities

TASK 4:

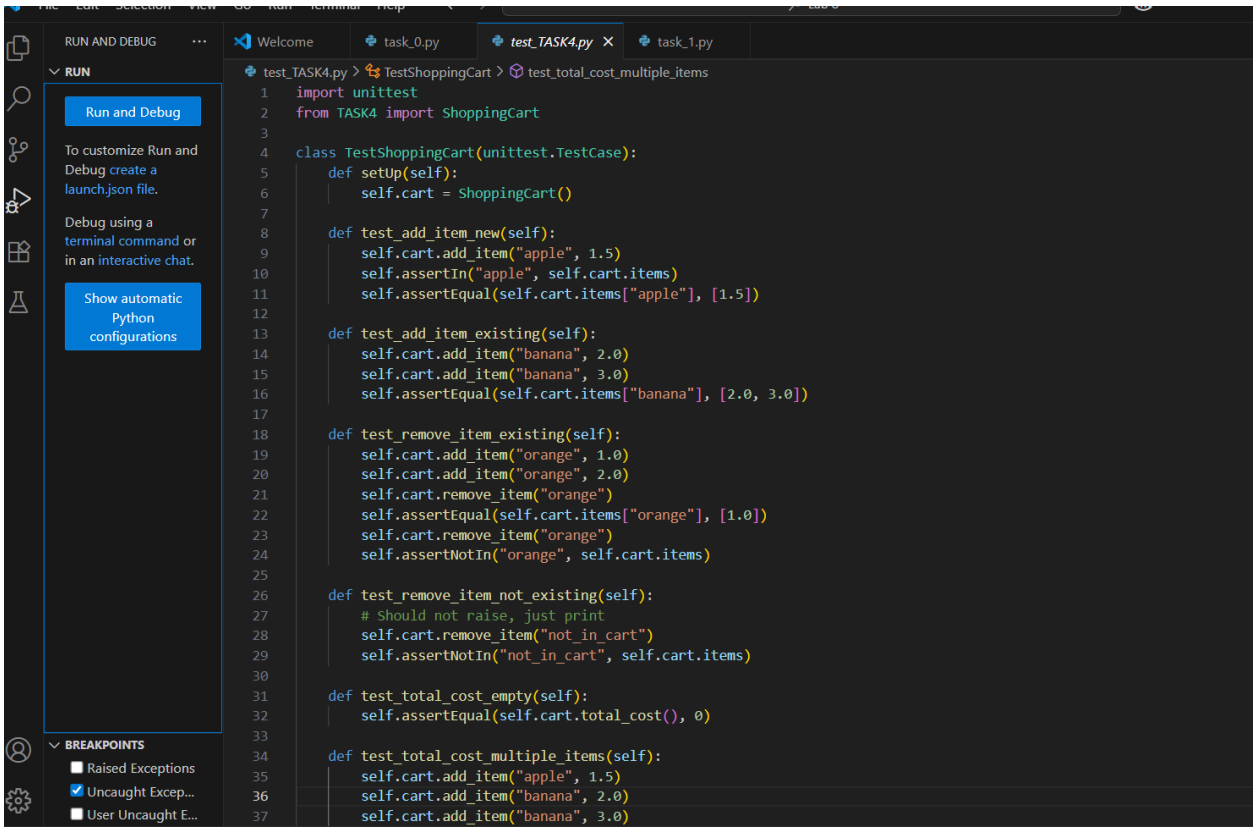


```python
class ShoppingCart:
    def __init__(self):
        self.items = {}

    def add_item(self, name, price):
        if name in self.items:
            self.items[name].append(price)
        else:
            self.items[name] = [price]

    def remove_item(self, name):
        if name in self.items:
            self.items[name].pop()
            if not self.items[name]:
                del self.items[name]
        else:
            print(f"Item '{name}' not found in cart.")

    def total_cost(self):
        return sum(price for prices in self.items.values() for price in prices)

if __name__ == "__main__":
    cart = ShoppingCart()
    while True:
        print("\nShopping Cart Menu:")
        print("1. Add item")
        print("2. Remove item")
        print("3. View total cost")
        print("4. Exit")
        choice = input("Enter your choice (1-4): ").strip()
        if choice == "1":
            name = input("Enter item name: ").strip()
            price_input = input("Enter item price: ").strip()
            try:
                price = float(price_input)
                if price < 0:
                    print("Price cannot be negative.")
                    continue
                cart.add_item(name, price)
                print(f"Added '{name}' with price {price:.2f} to cart.")
            except ValueError:
                print("Invalid price. Please enter a number.")
        elif choice == "2":
            name = input("Enter item name to remove: ").strip()
            cart.remove_item(name)
        elif choice == "3":
            total = cart.total_cost()
            print(f"Total cost of items in cart: {total:.2f}")
        elif choice == "4":
            print("Exiting Shopping Cart. Goodbye!")
            break
        else:
            print("Invalid choice. Please enter a number from 1 to 4.")
```

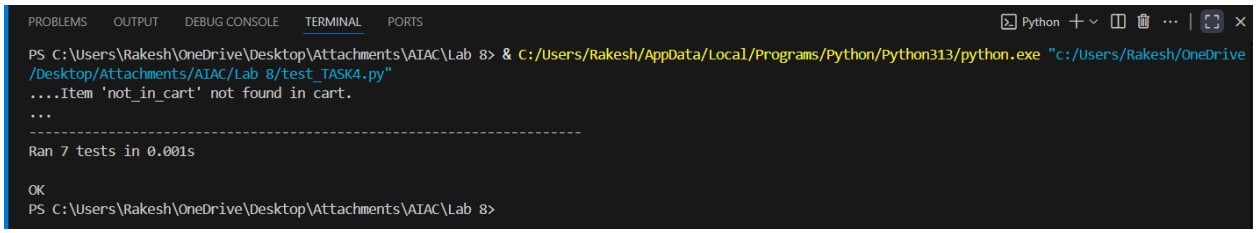
OUTPUT:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                   Python

blems (Ctrl+Shift+M)  kesh\OneDrive\Desktop\Attachments\AIAC\Lab 8> & C:/Users/Rakesh/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/Rakesh/OneDri
/Desktop/Attachments/AIAC/Lab 8/TASK4.py"

Shopping Cart Menu:
1. Add item
2. Remove item
3. View total cost
4. Exit
Enter your choice (1-4): 1
Enter item name: nuts
Enter item price: 190
Added 'nuts' with price 190.00 to cart.

Shopping Cart Menu:
1. Add item
2. Remove item
3. View total cost
4. Exit
Enter your choice (1-4): 3
Total cost of items in cart: 190.00

Shopping Cart Menu:
1. Add item
2. Remove item
3. View total cost
4. Exit
Enter your choice (1-4): 2
Enter item name to remove: nuts

Shopping Cart Menu:
1. Add item
2. Remove item
3. View total cost
4. Exit
Enter your choice (1-4): 4
Exiting Shopping Cart. Goodbye!
PS C:\Users\Rakesh\OneDrive\Desktop\Attachments\AIAC\Lab 8>
```

TEST CASE:



OUTPUT:



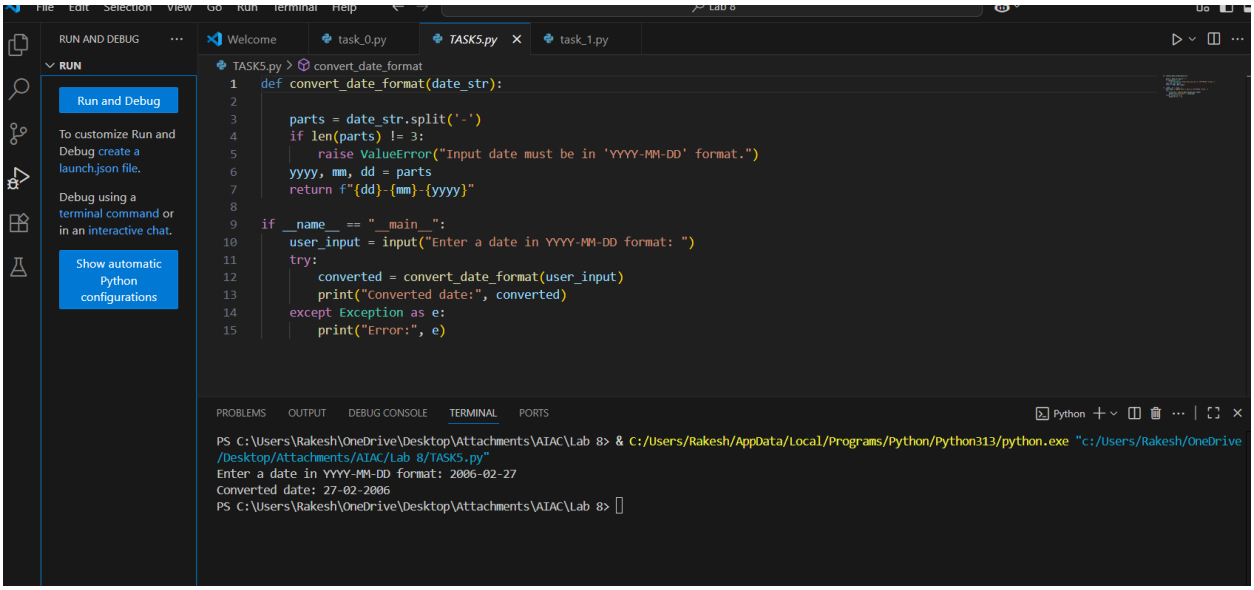
Task Description#5

- Use AI to write test cases for convert_date_format(date_str) to switch from "YYYY-MM-DD" to "DD-MM-YYYY".
  Example: "2023-10-15" → "15-10-2023"

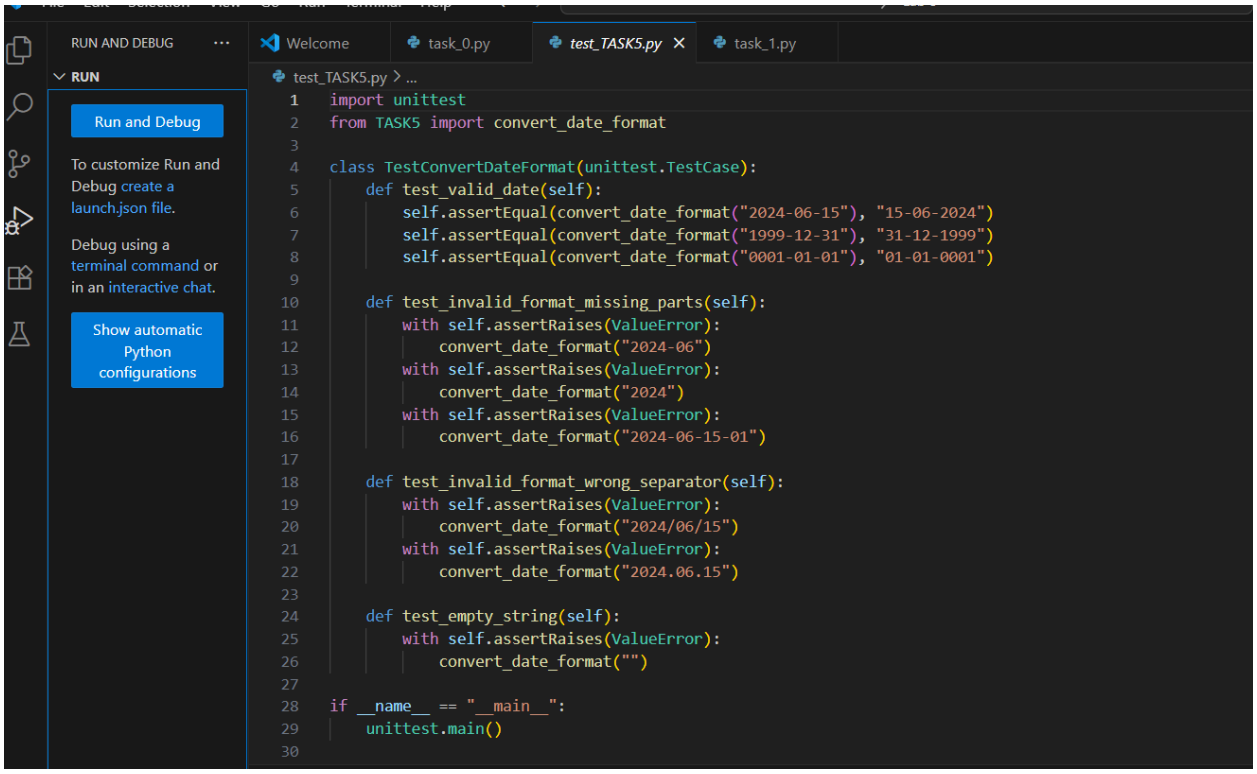
Expected Output#5

- Function converts input format correctly for all test cases

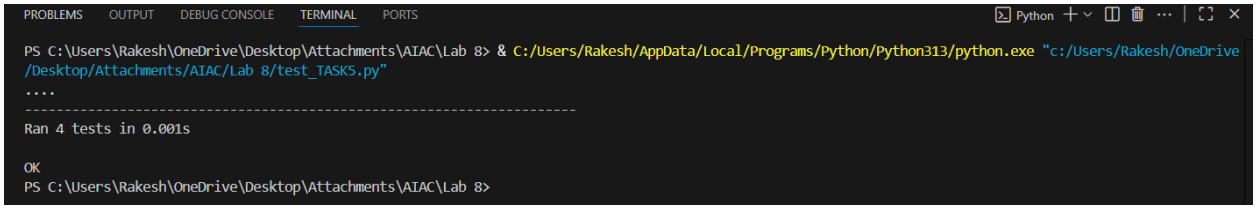
TASK 5:



TEST CASE:


RUN AND DEBUG
RUN
Run and Debug
To customize Run and Debug create a launch.json file.
Debug using a terminal command or in an interactive chat.
Show automatic Python configurations
Welcome
task_0.py
test_TASK5.py
task_1.py
test_TASK5.py > ...
1 import unittest
2 from TASK5 import convert_date_format
3
4 class TestConvertDateFormat(unittest.TestCase):
5 def test_valid_date(self):
6 self.assertEqual(convert_date_format("2024-06-15"), "15-06-2024")
7 self.assertEqual(convert_date_format("1999-12-31"), "31-12-1999")
8 self.assertEqual(convert_date_format("0001-01-01"), "01-01-0001")
9
10 def test_invalid_format_missing_parts(self):
11 with self.assertRaises(ValueError):
12 convert_date_format("2024-06")
13 with self.assertRaises(ValueError):
14 convert_date_format("2024")
15 with self.assertRaises(ValueError):
16 convert_date_format("2024-06-15-01")
17
18 def test_invalid_format_wrong_separator(self):
19 with self.assertRaises(ValueError):
20 convert_date_format("2024/06/15")
21 with self.assertRaises(ValueError):
22 convert_date_format("2024.06.15")
23
24 def test_empty_string(self):
25 with self.assertRaises(ValueError):
26 convert_date_format("")
27
28 if __name__ == "__main__":
29 unittest.main()
30

OUTPUT:


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\Rakesh\OneDrive\Desktop\Attachments\AIAC\Lab 8> & C:/Users/Rakesh/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/Rakesh/OneDrive/Desktop/Attachments/AIAC/Lab 8/test_TASK5.py"
....
Ran 4 tests in 0.001s
OK
PS C:\Users\Rakesh\OneDrive\Desktop\Attachments\AIAC\Lab 8>